🔹 Assignment 1: Permanent Table

Goal: Test persistence, Time Travel & recovery

Tasks:

1. Create a permanent table EMP_PERM
2. Insert minimum 5 records
3. Logout and login again
4. Verify data persists
5. Delete 2 records
6. Query deleted data using **Time Travel**
7. Restore deleted data
8. Drop the table
9. Recover using UNDROP TABLE

Learning Outcome:
✔ Data persists across sessions
✔ Supports Time Travel & Fail-safe

🔹 Assignment 2: Temporary Table

Goal: Test session-based background cleanup

Tasks:

1. Create temporary table EMP_TEMP
2. Insert records
3. Query data successfully
4. End the session
5. Reconnect and query the table

Learning Outcome:
✔ Table automatically removed after session ends
✔ No Time Travel or Fail-safe

🔹 Assignment 3: Transient Table

Goal: Test limited retention behavior

Tasks:

1. Create transient table EMP_TRANS

2. Insert records
3. Delete records
4. Query data using Time Travel (within retention)
5. Drop the table
6. Attempt recovery after retention period

Learning Outcome:
✓ Limited Time Travel
✓ No Fail-safe
✓ Faster background purge

🔹 Assignment 4: CTAS (Create Table As Select)

Goal: Test background data transformation

Tasks:

1. Create table EMP_HIGH_SALARY using CTAS
2. Filter salary > 70,000
3. Add calculated column SALARY_GRADE
4. Drop source table
5. Verify CTAS table still exists

Learning Outcome:
✓ Independent table creation
✓ Background transformation execution

🔹 Assignment 5: Table Cloning

Goal: Test zero-copy cloning

Tasks:

1. Clone EMP_PERM as EMP_CLONE
2. Update data in cloned table
3. Compare original vs clone
4. Explain storage behavior

Learning Outcome:
✓ Zero-copy storage
✓ Independent data modification

◆ Assignment 6: Time Travel Testing

Goal: Test point-in-time recovery

Tasks:

1. Perform DELETE on any table
2. Query using:
   - AT OFFSET
   - AT TIMESTAMP
3. Restore deleted data

Learning Outcome:
✓ Background versioning
✓ Point-in-time recovery

◆ Assignment 7: Drop & Purge Behavior

Goal: Compare cleanup rules

Tasks:

1. Drop Permanent table and recover it
2. Drop Transient table and attempt recovery
3. Drop Temporary table and verify immediate removal

Learning Outcome:
✓ Different purge behavior per table type

◆ Assignment 8: External Tables

Goal: Query data stored outside Snowflake

Tasks:

1. Create an external stage (S3 / ADLS / GCS)
2. Create an external table on CSV or JSON data
3. Query external table
4. Add new file to external storage
5. Refresh metadata and re-query

Questions to Answer:

- Does Snowflake store the data?
- Is Time Travel supported?

- What happens when source file is deleted?

Learning Outcome:
✓ Metadata-only tables
✓ No data storage inside Snowflake
✓ Manual metadata refresh

🔹 Assignment 9: Dynamic Tables

Goal: Test automated background refresh

Tasks:

1. Create a source table SALES_RAW
2. Insert sample sales data
3. Create a **Dynamic Table** SALES_AGG_DT
    - Aggregate total sales by region
4. Set refresh lag (e.g., 1 minute)
5. Insert new data into SALES_RAW
6. Observe automatic refresh

Questions to Answer:

- Is manual refresh required?
- How does refresh lag work?
- Difference between Dynamic Table and Task + Table?

Learning Outcome:
✓ Automatic background refresh
✓ Declarative transformation logic

🔹 Assignment 10: Comparison & Analysis

Goal: Conceptual understanding

Tasks:
Create a comparison table for:

- Permanent vs Temporary vs Transient
- Internal vs External Tables
- Static Tables vs Dynamic Tables

Include:

- Storage

- Time Travel
- Fail-safe
- Refresh behavior
- Use cases

📦 Student Deliverables

- SQL scripts
- Output screenshots
- Answers to conceptual questions
- Final comparison table

⏱️ Estimated Time

🕒 2 – 2.5 Hours

✅ Evaluation Criteria

- Correct execution: 40%
- Understanding of background behavior: 40%
- Explanation & documentation: 20%